



Supporting Information

3D Blockage Mapping for Identifying Familial Point Mutations in Single Amyloid- β Peptides with a Nanopore

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SUPPORTING INFORMATION

Table of Contents

Experimental Procedures.....	2
Table S1. Sequences, pI and net charges of the A β ₁₈₋₂₆ peptides.....	3
Supplementary Note 1 False Identification with Support Vector Machine (SVM).....	3
Table S2. The confusion matrix.....	3
Table S3. False identification of different parameter combination with Support Vector Machine (SVM).....	3
Figure S1. Mass Spectra of the WT and mutant A β ₁₈₋₂₆ peptides.....	4
Figure S2. <i>I</i> - <i>V</i> curves of the WT and mutant T232K/K238Q aerolysin nanopores.....	4
Figure S3. Current traces of the Azo-ODN detected with the WT aerolysin nanopore.....	4
Figure S4. The <i>I</i> _b / <i>I</i> ₀ , log(<i>t</i> _D), and σ_b histograms of typical blockade events for the Azo-ODN isomers.....	5
Figure S5. Detection of the WT and A21G A β ₁₈₋₂₆ peptides with the WT aerolysin nanopore.....	5
Figure S6. The voltage-dependent studies of the WT and A21G A β ₁₈₋₂₆ peptides.....	5
Figure S7. Current traces of the WT and A21G A β ₁₈₋₂₆ peptides measured with the T232K/K238Q aerolysin nanopore.....	6
Figure S8. The <i>I</i> _b / <i>I</i> ₀ , log(<i>t</i> _D), and σ_b histograms of typical blockade events for the WT and A21G A β ₁₈₋₂₆ peptides.....	6
Figure S9. The relative proportions of the WT and A21G A β ₁₈₋₂₆ peptides in the mixtures.....	6
Figure S10. The representative blockade events for the WT, A21G, E22Q and E22G A β ₁₈₋₂₆ peptides.....	6
Figure S11. The 3D scatter plot of the WT, A21G, E22Q and E22G A β ₁₈₋₂₆ mixture.....	7
References.....	7
Author Contributions.....	7

Experimental Procedures

1. Chemicals

Tris (hydroxymethyl) aminomethane (Tris, $\geq 99\%$), Ethylenediaminetetraacetic acid (EDTA, $\geq 99.99\%$), hexadecane, and hexane were purchased from Aladdin Co., Ltd. (Shanghai, China). KCl ($\geq 99\%$), decane (anhydrous, $\geq 99.8\%$) and trypsin-EDTA (0.25%) were purchased from Sigma Aldrich Co., Ltd. (St. Louis, MO, USA). 1,2-Diphytanoyl-sn-glycero-3-phosphocholine (DPhPC, powder, $\geq 99\%$) was purchased from Avanti Polar Lipids, Inc. (Alabama, USA), which was dissolved in decane to obtain the lipid solution with a concentration of 30.0 mg/mL. Hexadecane solution was prepared by mixing hexadecane and with a volume concentration ratio of 1:99 to facilitate the formation of lipid bilayer. The proaerolysin gene, primers, and A β ₁₈₋₂₆ peptides were synthesized and purified by GL Biochem Ltd (Shanghai, China). Azo-ODN were synthesized and purified by Sangon Biotech Co., Ltd (Shanghai, China).^[1] The proaerolysins were prepared in our laboratory according to the previous studies.^[2] To obtain the activated monomers, the proaerolysin was treated by trypsin-EDTA solution with a mass concentration ratio of 10:1 for 6 hours at room temperature. Unless otherwise indicated, the reagents and chemicals are all analytical grade. The solutions in our experiment were prepared with ultrapure water (18.2 M Ω ·cm at 25 °C) from an Arium Pro (Sartorius, USA).

2. Nanopore Experiments

Experimental apparatus. Nanopore detection was performed by using a home-made Teflon testing cell, which is composed by a *cis* and a *trans* chambers that clamped a 25- μ m thick Teflon sheet (FP301200, Goodfellow Cambridge Ltd., UK) in middle to separate the two components. Especially, in the Teflon sheets, an aperture of ~ 40 μ m in diameter was fabricated by a home-made electric spark generator. Voltages are applied through a pair of Ag/AgCl electrodes that were immersed in the electrolyte solutions of each chamber. The data of A β peptides were collected by an Axon 200B device digitized by 1440B digitizer (Axon Instruments), and the data of Azo-ODN were collected by the self-developed named "Cube". All data were acquired with a low-pass filter at 5 kHz and a sampling frequency of 100 kHz. The nanopore experiments in this work were all carried out at 24 ± 2 °C.

Single-channel Recording. Nanopore experiments were conducted by starting with the construction of a lipid bilayer in an appropriate thickness for aerolysin assembly. According to the Montal-Mueller method, the area around the aperture in the Teflon sheets is pretreated with hexadecane solution. After that, add 1.5-2.0 μ L lipid solution onto the surface of buffer solution in both the *cis* and *trans* chambers, and wait 10 minutes for the formation of a lipid monolayer. Then, moving the air-solution interface below and above the aperture with a syringe device, during which the lipid bilayer membrane would form due to the apposition of hydrocarbon chains. Next,

SUPPORTING INFORMATION

the activated proaerolysin solution was added into the *cis* chamber (connected to “ground”), with an external bias voltage of +200 mV applied to assist the self-assembly of aerolysin protein nanopore in the lipid bilayer. After insertion of a single aerolysin nanopore, the analytes were also added to the *cis* chamber, which induced typical transient current blockades.

3. Data analysis

In the nanopore analysis, the current blockade signals were extracted and analyzed by the lab-developed software “PyNanoLab” (<https://www.pynanolab.com>). In the statistical analysis, the events with duration time less than 0.20 ms were excluded for minimizing the evaluation errors. The mean values of the relative current blockage (I_b/I_0) and standard deviation of current blockade fluctuations (σ_b) were extracted from the Gaussian fits of the I_b/I_0 histogram and σ_b histogram, and the duration time (t_D) was obtained by the Gaussian fitting of the $\log(t_D)$ histogram. The average values of I_b/I_0 , t_D , and σ_b are shown as mean $\pm$ s. d., where the error bars were obtained from at least three independent experiments.

Supplementary Notes, Tables and Figures

Table S1. Sequences, pI and net charges of the A β_{18-26} peptides.

Peptides	Sequences	pI ^[a]	Net Charges ^[a]
WT A β_{18-26}	N'-VFFAEDVGS-C'	3.70	-2.78
A21G A β_{18-26}	N'-VFFGEDVGS-C'	3.70	-2.78
E22Q A β_{18-26}	N'-VFFAQDVGS-C'	3.84	-1.78
E22G A β_{18-26}	N'-VFFAGDVGS-C'	3.84	-1.78

^[a] The pI and net charge at pH 8.0 were calculated by the Peptide Tool (<https://www.protpi.ch/Calculator/PeptideTool>).

Supplementary Note 1 False Identification with Support Vector Machine (SVM)

The supervised Support Vector Machine (SVM) is used to classify the population in a mixed sample which is compromised with every 5000 events of pure WT and A21G A β_{18-26} sample. The trained SVM acts as a linear classifier and was evaluated by 10-fold cross-validation. For the binary classifiers, the false identification here was defined as the misclassification rate in SVM classification, which is carried out by using the confusion matrix as follows:

Table S2. The confusion matrix

		Predicted class	
		Positive	Negative
Actual class	Positive	True Positive (TP)	False Negative (FN)
	Negative	False Positive (FP)	True Negative (TN)

As shown in the confusion matrix, there are two possible predicted classes positive and negative obtained from the SVM model. TP and TN indicate that the predicted conditions are consistent with the true conditions, while FN and FP indicate that the predicted conditions are inconsistent with the true conditions. Here the false identification rate refers to the proportion of the SVM misclassification, which could be calculated as the following equation:

$$\text{False identification rate} = \frac{FP+FN}{TP+FN+FP+TN}$$

The false identification of the SVM model from different parameter combinations is shown in Table S3.

Table S3. False identification of different parameter combination with Support Vector Machine (SVM)

Parameter Combination	False Identification(%)
I_b/I_0 vs. $\log(t_D)$	23.3
$\log(t_D)$ vs. σ_b	13.1
I_b/I_0 vs. σ_b	12.4

SUPPORTING INFORMATION

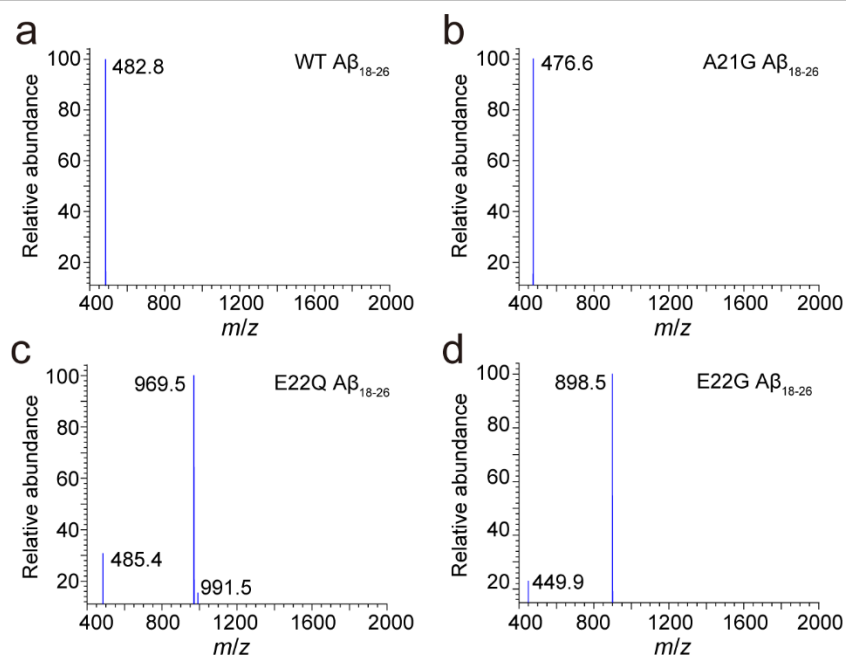


Figure S1. Mass Spectra of the (a) WT, (b) A21G, (c) E22Q, and (d) E22G $A\beta_{18-26}$ peptides, respectively.

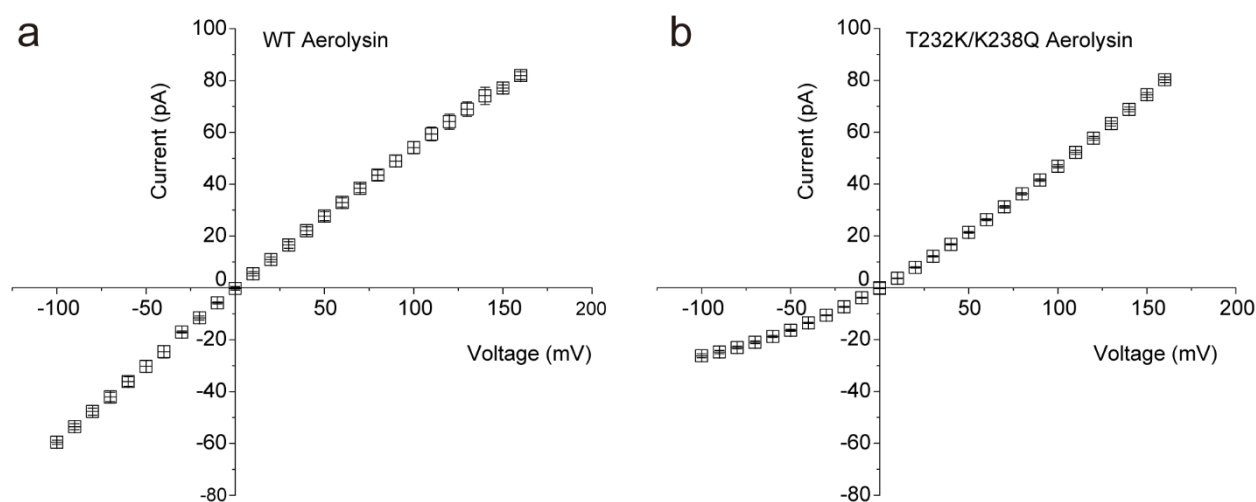


Figure S2. I - V curves of (a) the WT aerolysin nanopore in 1.0 M KCl, 10 mM Tris, 1.0 mM EDTA, pH 8.0, and (b) the mutant T232K/K238Q aerolysin nanopore in 1.0 M KCl, 10 mM Tris, pH 8.0, respectively.

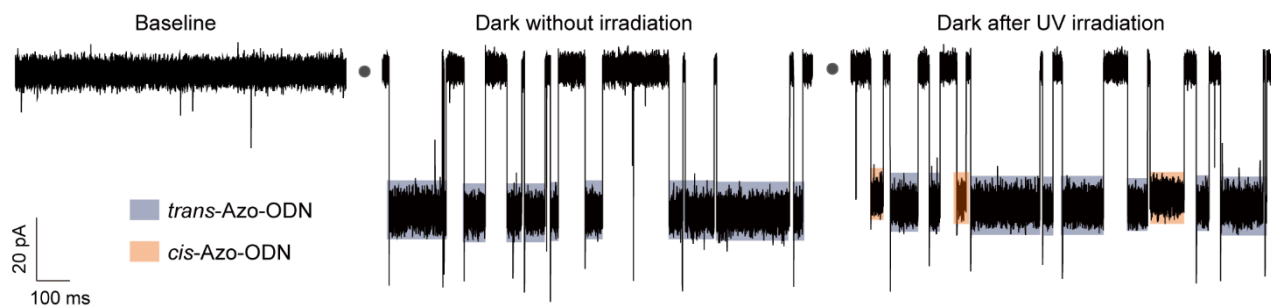


Figure S3. Current traces of the Azo-ODN detected by the WT aerolysin nanopore. From left to right: the baseline, the original current traces in the dark environment without and after UV irradiation ($\lambda = 365$ nm). All the data of Azo-ODN in this study were obtained using the WT aerolysin nanopore at +120 mV in 1.0 M KCl, 10 mM Tris, 1.0 mM EDTA, pH 8.0, respectively.

SUPPORTING INFORMATION

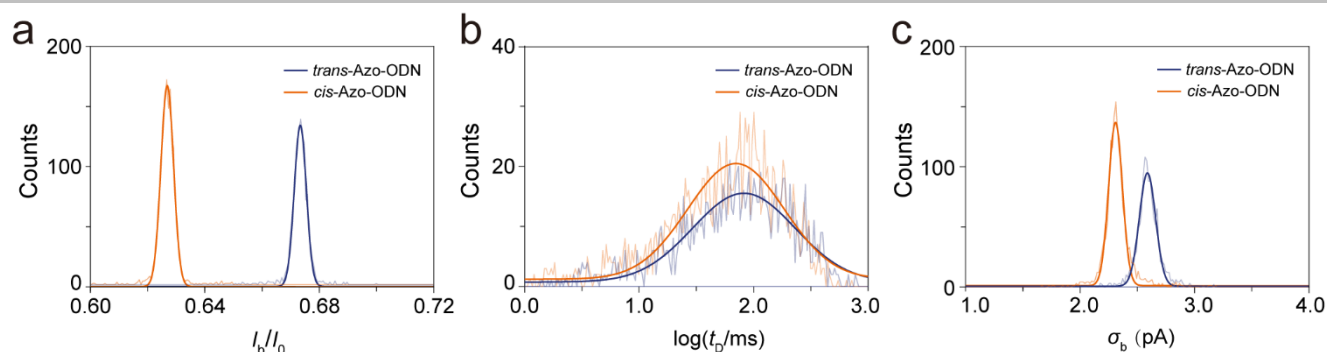


Figure S4. (a) I_b/I_0 , (b) $\log(t_D)$, and (c) σ_b histograms of the typical current blockade events for *cis*-Azo-ODN and *trans*-Azo-ODN.

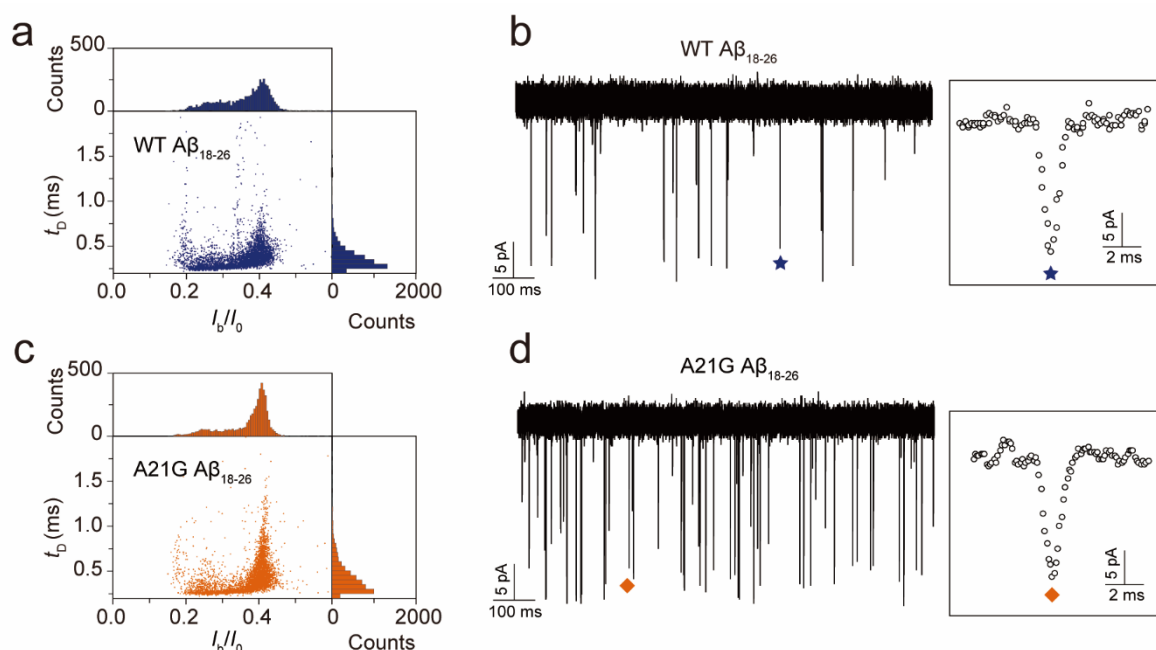


Figure S5. Detection of the WT and A21G $A\beta_{18-26}$ peptides with the WT aerolysin nanopore. The scatter diagrams, t_D histograms, and I_b/I_0 histograms for the WT (a) and the A21G (c) $A\beta_{18-26}$ peptides, respectively. The current trace and the typical current blockade events for the WT (b) and the A21G (d) $A\beta_{18-26}$ peptides, respectively. Each peptide was added to the *cis* chamber at a final concentration of 10.0 μ M. Data for $A\beta_{18-26}$ peptides were acquired at +120 mV in the buffer containing 1 M KCl and 10 mM Tris buffer, pH 8.0.

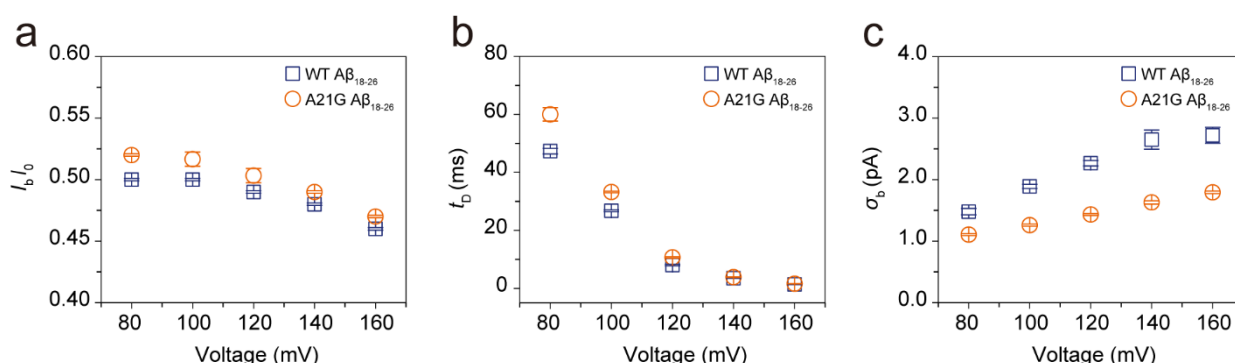


Figure S6. The voltage-dependent studies of the WT and A21G $A\beta_{18-26}$ peptides using the mutant T232K/K238Q aerolysin nanopore. Voltage dependencies of I_b/I_0 (a), t_D (b), and σ_b (c) for the WT and A21G $A\beta_{18-26}$ peptides with the applied voltages ranging from +80 to +160 mV, respectively. Each peptide was added to the *cis* chamber at a final concentration of 10.0 μ M. Data were acquired in the buffer containing 1 M KCl and 10 mM Tris buffer, pH 8.0.

SUPPORTING INFORMATION

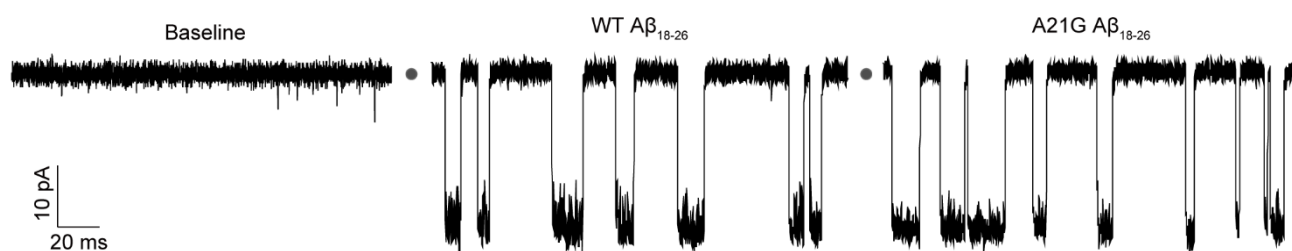


Figure S7. Current traces of the A β_{18-26} peptides monitored by the engineered T232K/K238Q aerolysin nanopore. From left to right: the baseline, the original current traces of the WT and A21G A β_{18-26} , respectively. Data of the peptides were recorded at +120 mV in 1.0 M KCl, 10 mM Tris, pH 8.0.

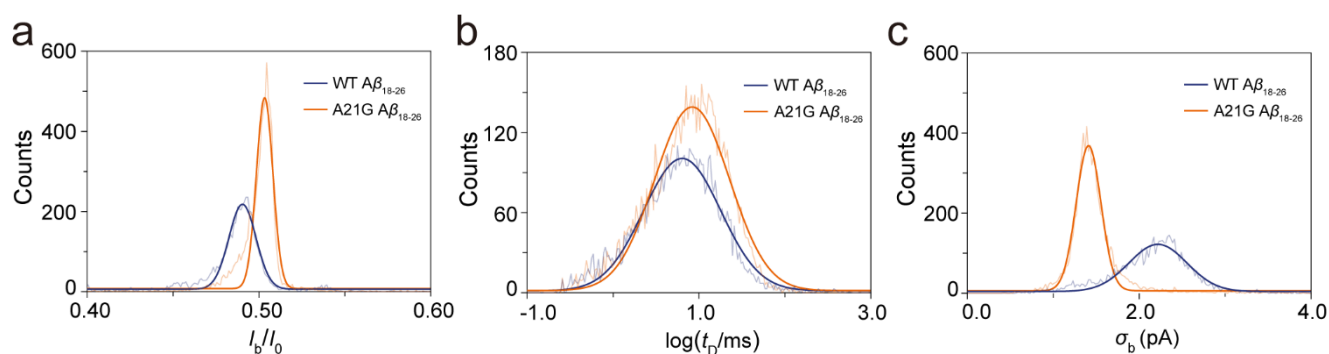


Figure S8. (a) I_b/I_0 , (b) $\log(t_D)$, and (c) σ_b histograms of the typical current blockade events for the WT and A21G A β_{18-26} peptides, respectively. Detection of the peptides was carried out by the engineered T232K/K238Q aerolysin nanopore.

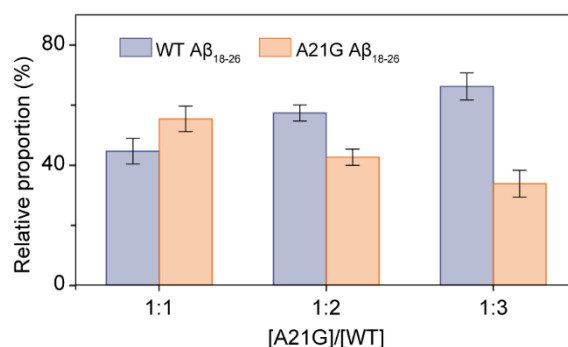


Figure S9. Relative proportions of peak areas for the WT and the A21G A β_{18-26} in the mixtures at different concentration ratios. The final concentration of the A21G A β_{18-26} was fixed at 5.0 μ M. The data were collected from a 5-minute recording of each mixture sample by using an engineered T232K/K238Q aerolysin nanopore.

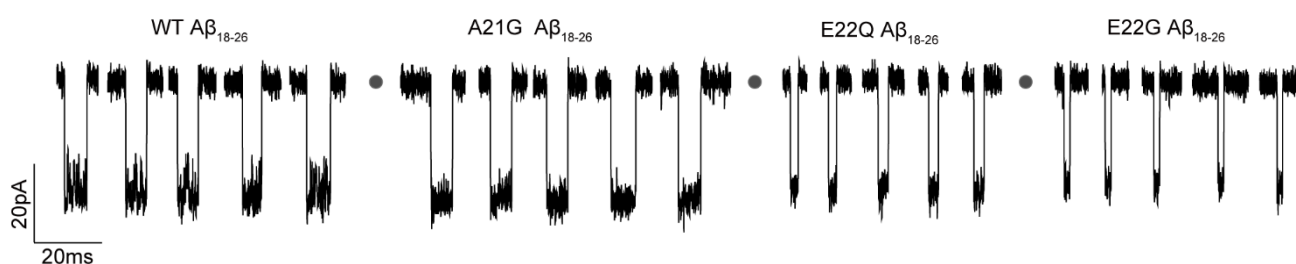


Figure S10. Representative blockade events of the WT, A21G, E22Q, E22G A β_{18-26} peptides measured by the engineered T232K/K238Q aerolysin nanopore (from left to right), respectively.

SUPPORTING INFORMATION

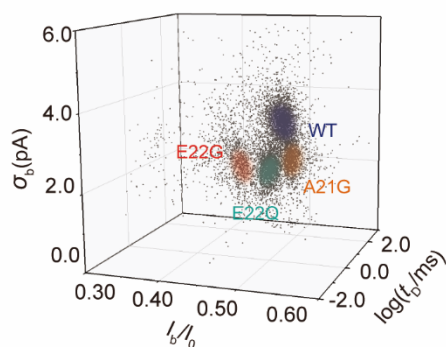


Figure S11. 3D scatter plot of I_b/I_0 vs. $\log(t_D)$ vs. σ_b for the mixture sample of the mutant A21G, E22Q, E22G and the WT $A\beta_{18-26}$ peptides. The mixture sample was detected by the engineered T232K/K238Q aerolysin nanopore.

References

- [1] Z.-L. Hu, Z.-Y. Li, Y.-L. Ying, J. Zhang, C. Cao, Y.-T. Long, H. Tian, *Anal. Chem.* **2018**, *90*, 4268-4272.
- [2] X.-Y. Wu, M.-B. Wang, Y.-Q. Wang, M.-Y. Li, Y.-L. Ying, J. Huang, Y.-T. Long, *CCS Chem.* **2019**, *1*, 304-312.

Author Contributions

Y.-T. Long and Z.-L. Hu guided this research project. Z.-L. Hu designed the experiments. K.-L. Xin conducted the experiments and collected the nanopore data. K.-L. Xin and Z.-L. Hu processed the data. K.-L. Xin, Z.-L. Hu, S.-C. Liu, X.-Y. Li, Y.-L. Ying and Y.-T. Long analyzed the data and discussed the results. S.-C. Liu developed the PyNanoLab software for nanopore data analysis. J.-G. Li prepared the proaerolysin protein. H. Niu, conducted the pre-experiments of $A\beta$ peptides. K.-L. Xin and Z.-L. Hu prepared the manuscript. Y.-L. Ying and Y.-T. Long revised and improved the manuscript.